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WES 267B

Assignment 4

Assignment 4 Report

4.1 blas_mm Lab

The table below highlights the lab results obtained by running the ``time make run`` command.

Measurement	Assignment #2	Assignment #3	Assignment #4 (B=1)	Assignment #4 (B=1,)
Real time	0m32.536s	0m4.310s	0m10.440s	0m9.970s
User time	0m32.176s	0m2.258s	0m9.183s	0m8.718s
System time	0m0.225s	0m0.785s	0m0.700s	0m0.890s

Measurement	3.1 OpenCL Naïve MM	3.2 OpenCL Tiled MM	3.3 Neon Block MM (256x256 Block)
Real time	0m 50.718s	0m4.310s	0m24.954s
User time	0m 2.154s	0m2.258s	0m24.512s
System time	0m 0.755s	0m0.785s	0m0.323s

Measurement	2.1 Naïve MM	2.2 Block MM	2.3 Block MM Unrolled
Real time	1m15.191s	0m34.149s	0m32.536s
User time	1m14.574s	0m33.819s	0m32.176s
System time	0m0.336s	0m0.215s	0m0.225s

	Speedup (x)
Assignment #2	15.72
Assignment #3	4.54

* Please use the ``time`` command to compare this against your implementations in Assignment 2 and 3. Is this faster for all matrices? If not, why?

Answer: Assignment #4 achieved a 3.12x speedup over Assignment #2. However, Assignment #3's OpenCL Tiled MM was 2.42x faster than Assignment #4. Therefore, Assignment #4 was not faster than all previous implementations. The OpenCL tiled

implementation was likely faster because it was specifically optimized for the tested matrix sizes and the Qualcomm Adreno 643 GPU. In comparison, CLBlast uses a more general implementation that was not specifically tuned for the Qualcomm Adreno 643 GPU. Since, the batch size was one, there was no performance benefit from the batched execution so that likely explains why it wasn't the fastest all around.

Lab Output (`time make run`)

```
ubuntu@9b593b7a4f1a: /workspaces/wes237b/assignment-4/lab/blas_ms$ time make run
g++ -g -Wall -o solution ../../helper_lib/helper_lib.a main.cc -I../../helper_lib -I/usr/local/cuda/include -I../../helper_lib/helper_lib.a -ln -lcblast -L/usr/local/cuda/lib64 -lOpenCL
In file included from /usr/include/CL/cl.h:20,
                 from /usr/include/CL/opencl.h:24,
                 from /usr/local/include/CLBlast.h:27,
                 from main.cc:5:
/usr/include/CL/cl_version.h:22:104: note: '#pragma message: cl_version.h: CL_TARGET_OPENCL_VERSION is not defined. Defaulting to 300 (OpenCL 3.0)'
22 | #pragma message("cl_version.h: CL_TARGET_OPENCL_VERSION is not defined. Defaulting to 300 (OpenCL 3.0)")
   | ~~~~~^
main.cc: In function 'void OpenCLMatrixMultiply(Matrix*, Matrix*, Matrix*)':
main.cc:103:23: warning: unused variable 'status' [-Wunused-variable]
103 |     cblast::StatusCode status = cblast::GemvBatched<float>(cblast::Layout::RowMajor, cblast::Transpose::No, cblast::Transpose::No, M, N, K, alphas, device_a, a_offsets, K, device_b, b_offsets, N, B, &queue, nullptr);
   |                      ^~~~~~
./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/output.raw output.raw
Running on:
Platform:  QUALCOMM Snapdragon(TM)
Device:    QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m11.179s
user    0m9.775s
sys     0m0.830s
o ubuntu@9b593b7a4f1a: /workspaces/wes237b/assignment-4/lab/blas_ms$
```

Lab Output (`make benchmark`)

I modified the Makefile to run `time` for each dataset folder.

Assignment 4	Assignment 3	Assignment 2
<pre>ubuntu@593b7a71ac:/workspaces/wes237b/assignment-4/lab/blas_m4\$ make benchmark time ./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.788s user 0m0.785s sys 0m0.675s time ./solution Dataset/1/input1.raw Dataset/1/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.744s user 0m0.694s sys 0m0.638s time ./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.753s user 0m0.724s sys 0m0.654s time ./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.753s user 0m0.788s sys 0m0.688s time ./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.749s user 0m0.781s sys 0m0.636s time ./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.768s user 0m0.655s sys 0m0.651s time ./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.779s user 0m0.718s sys 0m0.656s time ./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.754s user 0m0.788s sys 0m0.641s time ./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.751s user 0m0.783s sys 0m0.655s time ./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m0.759s user 0m0.694s sys 0m0.651s time ./solution Dataset/10/input0.raw Dataset/10/output.raw output.raw Running on: Platform: QUILCOM Snapdragon(TM) Device: QUILCOM Adreno(TM) 643 !!SOLUTION IS CORRECT!! real 0m2.759s user 0m2.137s sys 0m0.189s ubuntu@593b7a71ac:/workspaces/wes237b/assignment-4/lab/blas_m4\$</pre>	<pre>root@e3d434739bb3:/workspaces/wes237b/assignment-3/lab/ncnn\$ make benchmark time ./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw sum: 129 == 129 !!SOLUTION IS CORRECT!! real 0m0.003s user 0m0.000s sys 0m0.002s time ./solution Dataset/1/input0.raw Dataset/1/output.raw output.raw sum: -11 == -11 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw sum: 143 == 143 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.002s sys 0m0.000s time ./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw sum: 305 == 305 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw sum: -270 == -270 !!SOLUTION IS CORRECT!! real 0m0.004s user 0m0.001s sys 0m0.003s time ./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw sum: -46 == -46 !!SOLUTION IS CORRECT!! real 0m0.004s user 0m0.002s sys 0m0.002s time ./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw sum: 986 == 986 !!SOLUTION IS CORRECT!! real 0m0.004s user 0m0.000s sys 0m0.004s time ./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw sum: 760 == 760 !!SOLUTION IS CORRECT!! real 0m0.004s user 0m0.001s sys 0m0.003s time ./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw sum: 76 == 76 !!SOLUTION IS CORRECT!! real 0m0.003s user 0m0.000s sys 0m0.003s time ./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw sum: 778 == 778 !!SOLUTION IS CORRECT!! real 0m0.003s user 0m0.001s sys 0m0.002s root@e3d434739bb3:/workspaces/wes237b/assignment-3/lab/ncnn\$</pre>	<pre>ubuntu@f2d7b4ab1bec:/workspaces/wes237b/assignment-2/lab/loop_unrollings\$ make benchmark gcc -g -Wall -O3 solution ../helper_lib/helper_lib.a main.c -I../helper_lib -I/usr/lo cal/cuda/include -I../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL time ./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw sum: 129 == 129 !!SOLUTION IS CORRECT!! real 0m0.005s user 0m0.000s sys 0m0.004s time ./solution Dataset/1/input0.raw Dataset/1/output.raw output.raw sum: -11 == -11 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw sum: 143 == 143 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw sum: 305 == 305 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw sum: -270 == -270 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw sum: -46 == -46 !!SOLUTION IS CORRECT!! real 0m0.004s user 0m0.001s sys 0m0.003s time ./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw sum: 986 == 986 !!SOLUTION IS CORRECT!! real 0m0.003s user 0m0.000s sys 0m0.003s time ./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw sum: 760 == 760 !!SOLUTION IS CORRECT!! real 0m0.004s user 0m0.001s sys 0m0.002s time ./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw sum: 76 == 76 !!SOLUTION IS CORRECT!! real 0m0.002s user 0m0.000s sys 0m0.002s time ./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw sum: 778 == 778 !!SOLUTION IS CORRECT!! real 0m0.003s user 0m0.000s sys 0m0.002s ubuntu@f2d7b4ab1bec:/workspaces/wes237b/assignment-2/lab/loop_unrollings\$</pre>

4.2 convolution-2d Lab Report

Describe how you implemented the convolution operation, any optimizations you made, and the performance characteristics of your solution.

I implemented the 2D convolution using OpenCL, where each GPU work-item computes one output pixel for one image channel. The kernel applies the mask over the input image using the specified stride and performs VALID convolution i.e. ensuring that the filter mask remains completely within the boundaries of the 2D matrix.

For optimization, I tested various local work group sizes (see chart below). The global work size is padded to a multiple of the local size, with bounds checking for extra work items. This was needed for the autograder. Otherwise, hardcoding the local work size to be a static number was passing test locally but failed running on the autograder. I also created a new kernel opencl file that explicitly used the local variable ensuring shared memory instead of pulling from the global memory for each convolution multiplication (see results titled shared memory.)

4.2 convolution-2d Lab Output (Global)

Measurement	4.2 convolution-2d			
Local work size	4x4x1	8x8x1	16x16x1	32x32x1
Real time	0m7.103s	0m6.876s	0m6.999s	0m6.879s
User time	0m4.571s	0m4.570s	0m4.622s	0m4.602s
System time	0m0.774s	0m0.739s	0m0.719s	0m0.752s

From left to right global shared memory 4x4x1, 8x8x1, 16x16x1, 32x32x1

4.2 convolution-2d Lab Output (Shared Memory)

I created a kernel opencl file to use local variable to ensure shared memory instead of pulling from the global memory for each convolution multiplication. The main issue is doing this way requires the use of barrier(CLK_LOCAL_MEM_FENCE) which is needed to ensure all work-items in the work-group have completed their local memory writes before any work-item begins using that shared data for the convolution calculations.

The tile size is dynamically determined using the below:

```
size_t tileWidth = (localWorkSize[0] - 1) * stride + maskWidth;
size_t tileHeight = (localWorkSize[1] - 1) * stride + maskWidth;
size_t tileSize = tileWidth * tileHeight;
```

Measurement	4.2 convolution-2d			
Local work size	4x4x1	8x8x1	16x16x1	32x32x1
Real time	0m7.572s	0m6.397s	0m7.114s	0m7.177s
User time	0m4.698s	0m4.717s	0m4.795s	0m4.826s
System time	0m0.778s	0m0.734s	0m0.710s	0m0.756s

From left to right 4x4x1, 8x8x1, 16x16x1, 32x32x1

I optimized the convolution using tiled local memory. Each work-group tested loads the input tile into local memory, allowing the work-items to reuse the data instead of repeatedly accessing global memory. This improves the efficiency of the convolution kernel. However, the overall application runtime showed little improvement.

4.2.2 With strides output (global shared memory)

```
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
!!SOLUTION IS CORRECT!!
Running test 36
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
Running test 37
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m12.822s
user    0m8.784s
sys     0m1.403s
○ ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$
```

Global shared memory 4x4x1

```
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
!!SOLUTION IS CORRECT!!
Running test 36
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
Running test 37
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m12.513s
user    0m8.703s
sys     0m1.374s
○ ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$
```

Global shared memory 8x8x1

```

ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
!!SOLUTION IS CORRECT!!
Running test 36
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
Running test 37
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m12.661s
user    0m8.679s
sys     0m1.350s
○ ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ █

```

Global shared memory 16x16x1

```

ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
!!SOLUTION IS CORRECT!!
Running test 36
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
Running test 37
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m12.678s
user    0m8.800s
sys     0m1.404s
○ ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ █

```

Global shared memory 32x32x1

4.2.3 With strides (local memory)

```
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
clSetKernelArg 8 failed: -49
Running test 36
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
Running test 37
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
make: *** [Makefile:34: with_stride] Error 1

real    0m8.487s
user    0m7.103s
sys     0m1.164s
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$
```

Local memory 4x4x1

```
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
clSetKernelArg 8 failed: -49
Running test 36
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
Running test 37
Running on:
  Platform: QUALCOMM Snapdragon(TM)
  Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
make: *** [Makefile:34: with_stride] Error 1

real    0m8.452s
user    0m7.046s
sys     0m1.190s
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$
```

Local memory 8x8x1

```

ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
clSetKernelArg 8 failed: -49
Running test 36
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
Running test 37
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
make: *** [Makefile:34: with_stride] Error 1

real    0m8.531s
user    0m7.059s
sys     0m1.247s
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$

```

Local memory 16x16x1

```

ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$ time make with_stride
clSetKernelArg 8 failed: -49
Running test 36
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
Running test 37
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

clSetKernelArg 8 failed: -49
make: *** [Makefile:34: with_stride] Error 1

real    0m8.462s
user    0m7.047s
sys     0m1.207s
ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/lab/convolution-2d$

```

Local memory 32x32x1

4.3 Homework CNN Output

```
● ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/homework/cnn$ make cpu
./m1 100
Test batch size: 100
Could not find a CPU or other requested device. Defaulting to first available device...
Loading fashion-mnist data...Done
Loading model...Done
Conv-OpenCL==
Layer Time: 906.63 ms
Op Time: 786.297 ms
Conv-OpenCL==
Layer Time: 792.465 ms
Op Time: 733.399 ms

Test Accuracy: 0.86

● ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/homework/cnn$ make gpu
./m2 100
Test batch size: 100
Running on:
    Platform: QUALCOMM Snapdragon(TM)
    Device: QUALCOMM Adreno(TM) 643

Loading fashion-mnist data...Done
Loading model...Done
Conv-OpenCL==
Layer Time: 894.102 ms
Op Time: 777.453 ms
Conv-OpenCL==
Layer Time: 769.56 ms
Op Time: 715.488 ms

Test Accuracy: 0.86

○ ubuntu@9b593b7a4f1a:/workspaces/wes237b/assignment-4/homework/cnn$
```